

Assignment-2

The objective of this assignment is to understand and implement the **Naive Bayes algorithm for text classification**. Students will learn how text documents are represented as numerical features and how Naive Bayes can be used to classify documents into predefined categories such as **spam/ham, positive/negative sentiment, or news categories**.

Dataset

Implement a sentiment-classification experiment using a dataset such as the IMDb movie-review dataset.

Required preprocessing

Perform the following preprocessing operations:

1. Convert text to lowercase.
2. Remove punctuation.
3. Remove unnecessary special characters.
4. Tokenize the text.
5. Remove stopwords.
6. Apply stemming or lemmatization.
7. Convert the processed documents into numerical feature vectors using **Bag-of-Words**

Do following tasks for Naive Bayes and Multinomial Naive Bayes classification.

Task 1: Data Exploration

Load the dataset and answer the following questions:

1. How many documents are present in the dataset?
2. How many documents belong to each class?
3. What is the average document length?
4. Which are the 20 most frequent words in the complete dataset?

Task 2: Text Preprocessing

Implement a preprocessing pipeline that performs:

- Lowercase conversion
- Tokenization
- Punctuation removal
- Stopword removal
- Lemmatization/stemming

Compare the dataset **before and after preprocessing**.

Task 3: Feature Extraction

Represent the documents using: Bag of words

Task 4: Naive Bayes Classification

Train a **Multinomial Naive Bayes and Binary Naive classifier** using the training data.

Perform the following:

1. Train the model.
2. Predict the class labels for the test set.
3. Calculate accuracy.
4. Calculate precision, recall, and F1-score.
5. Generate a confusion matrix.

Task 5: Model Comparison

Compare the performance of different models.

Task 6: Effect of Smoothing

Task 7: Error Analysis

Identify at least **10 incorrectly classified documents**.

For each incorrectly classified document, provide:

- Actual class
- Predicted class
- Important words/features
- Possible reason for misclassification

Report the accuracy, precision, recall, and F1-score of both the classifiers and explain which model performs best and why.